When Attention Becomes Invisible - and When It Helps: ECA, Supervision Realism, and Downstream ESPI Denoising

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Revision draft prepared from the reviewer audit and corrected experiments

Abstract. Efficient Channel Attention (ECA) is commonly added to convolutional denoisers without verifying that its internal channel modulation survives normalization and improves the downstream scientific task. We study this question in Electronic Speckle Pattern Interferometry (ESPI), where denoising is followed by five-class vibration-mode recognition. A protocol audit showed that the earlier three-run robustness summary did not establish independent seed propagation; we therefore replace it with a corrected five-seed end-to-end replication. In the random in-distribution stress protocol, aggressive V5R ECA reached Macro-F1 0.8589 +/- 0.0314, compared with 0.8269 +/- 0.0227 for light V4R and 0.8439 +/- 0.0164 for Raw. The paired V5R-minus-V4R estimate was +0.0319 (95% CI [0.0085, 0.0554]), but the exact sign-flip p-value was 0.0625 at n=5. In a stricter six-board transfer audit, Raw remained strongest in mean Macro-F1 (0.4970), showing that denoising is not universally beneficial. A matched residual U-Net-Lite ablation further showed that ECA utility depends on normalization: at epoch 15, GroupNorm yielded a +0.1429 board-balanced Macro-F1 effect, whereas BatchNorm yielded -0.0144. Gate interventions and affine-parameter audits reject a universal BatchNorm-absorption explanation; GroupNorm produced nearly static useful channel scaling, while BatchNorm propagated more dynamic but less useful modulation. The results support a conditional claim: supervision realism, normalization, topology, and evaluation protocol jointly determine whether ECA helps downstream ESPI classification.

Keywords: ESPI, denoising, Efficient Channel Attention, DnCNN, U-Net, normalization, reproducibility, downstream classification

1 Introduction

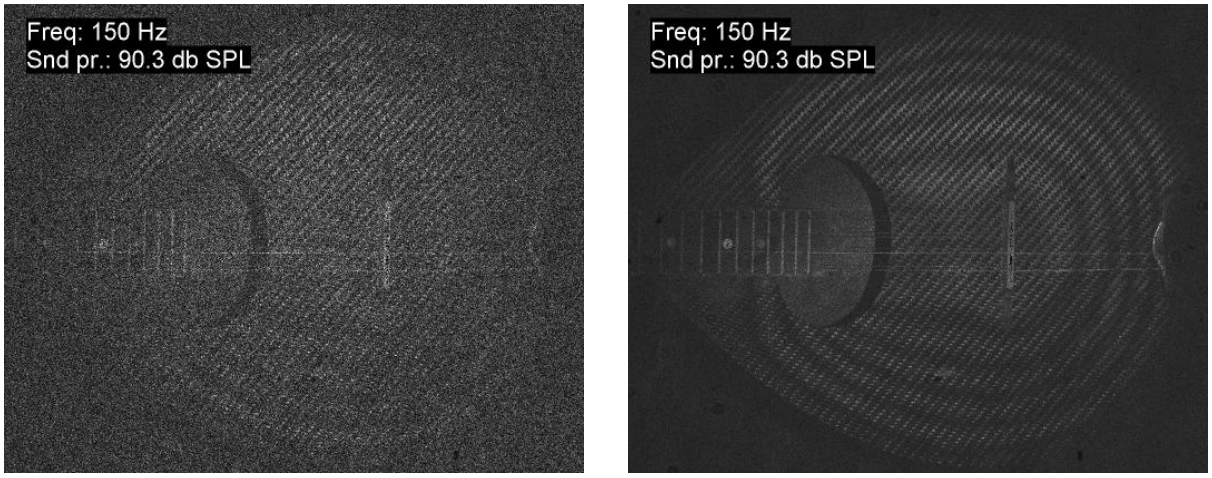
Electronic Speckle Pattern Interferometry (ESPI) is an optical full-field measurement technique. When a vibrating surface is illuminated coherently, deformation changes the interference pattern recorded by the camera. The resulting fringe field encodes modal structure: fringe spacing and curvature, nodal regions, and their spatial organization vary with the vibration state and material. A single-shot frame is also dominated by speckle and acquisition noise. Consequently, denoising can help only if it suppresses nuisance variation while preserving the structures used by the downstream classifier.

Channel attention is a plausible low-cost mechanism for this task. ECA computes a channel descriptor by global average pooling, applies a local one-dimensional convolution across channels, and multiplies each feature map by the resulting gate. Yet architectural presence is not evidence of functional value. Normalization, residual prediction, following convolutions, and the supervision target can attenuate or redirect the modulation. A denoiser may improve a reconstruction loss while erasing weak class-discriminative fringe evidence.

The original short paper reported float-level ECA/NoECA equivalence in a pseudo-supervised BatchNorm DnCNN configuration and a preliminary three-run stress summary. The reviewer audit motivated four corrections: explain the ESPI domain, replace the unverified robustness summary with a seed-aware replication, test ECA in a U-Net backbone, and directly inspect the proposed absorption mechanism. This revision makes those changes and narrows the claim wherever the corrected evidence does not support a universal conclusion.

- A corrected five-seed end-to-end DnCNN replication with explicit random, NumPy, PyTorch CPU, CUDA, split, sampler, and stress-noise seed control.
- A locked leave-one-board-out audit that separates random in-distribution robustness from transfer to unseen physical boards.
- A matched residual U-Net-Lite ECA ablation under GroupNorm and BatchNorm.
- A disjoint calibration/audit intervention study of ECA gates, normalization affine parameters, and downstream consumer paths.

2 ESPI data and downstream task



(a) Real single-shot input **(b) Averaged proxy target**

Figure 1. Domain bridge for non-specialists. A single-shot ESPI frame contains strong speckle and acquisition noise, while the aligned repeated-acquisition proxy is much smoother and can become nearly white. The classifier depends on spatial fringe spacing, curvature, nodal organization, and their material-dependent distribution, not on visual smoothness alone. Example: carbon board C01, 150 Hz, approximately 90 dB SPL.

The read-only acquisition inventory contains six physical boards: carbon C01-C03 and wood W01-W03. It contains 1,413 excitation-condition folders, 23,769 single-shot PNG acquisitions, and 1,413 supplied averaged-proxy targets. The number of available single shots varies by condition; the filesystem audit found a median of 16 and a range of 1-38. We do not infer that every repeat contributed to every supplied average because the original averaging-generation source was not available in the denoising repository.

The denoiser cache contains 23,891 aligned input/target pairs: 23,648 real single-shot inputs aligned to averaged proxies and 243 W01 PseudoNoisy inputs aligned to the same type of averaged proxy. We therefore describe this regime as averaged-proxy or predominantly real-aligned supervision, not as an exclusively real-single-shot dataset. The five-class classifier inventory contains 13,278 labeled images, with

strong class imbalance; the largest class contains 10,748 images and the smallest 267. Macro-F1 is therefore the primary downstream endpoint.

Table 1. Data and protocol inventory used in the corrected analyses.

Component	Locked detail
Physical boards	C01-C03 carbon; W01-W03 wood
Acquisition inventory	23,769 single shots; 1,413 averaged proxies
Denoiser pair cache	23,891 pairs: 23,648 real-aligned + 243 PseudoNoisy
Denoiser split	90% train / 10% validation; seed-controlled; 21,502 / 2,389
Classifier inventory	13,278 labeled grayscale images; five classes
Random stress split	Stratified 80% train / 20% validation per seed
Grouped transfer	11,706 images per fold; four train boards, one validation board, one test board

3 Models and training

3.1 DnCNN-Lite

The DnCNN-Lite denoiser has 17 convolutional layers, 32 feature channels, residual-noise prediction, and GroupNorm with eight groups. V4R inserts ECA at blocks [0, 1, 2]; V5R inserts ECA at [0, 1, 2, 3, 6, 10, 14]. Both use kernel size 5 and averaged-proxy supervision. Each of the five runs trains for 20 epochs with 256 x 256 crops, batch size 8, AdamW (learning rate 2e-4, weight decay 1e-5), cosine annealing, gradient clipping at 1.0, and a 0.7 Charbonnier + 0.3 MSE objective. Validation is never augmented, and the minimum validation-loss checkpoint is passed to the downstream evaluator.

3.2 Residual U-Net-Lite

To address backbone narrowness, we implement a parameter-controlled residual U-Net-Lite with encoder widths 8, 16, and 32, a 64-channel bottleneck, bilinear upsampling, skip concatenation, and two 3 x 3 convolutions per stage. The output predicts residual noise. ECA3 is placed after encoder stages enc0, enc1, and enc2. Its centered multiplier is constrained around identity with kernel size 5, temperature 1.0, and gain 0.25. The matched 2 x 2 comparison crosses ECA3 versus NoECA with GroupNorm (four groups) versus BatchNorm2d. All arms use the same seed-42 split manifest, 20 epochs, 256 x 256 crops, batch size 8, AdamW (2e-4, weight decay 1e-5), cosine annealing, mixed precision, gradient clipping, and the same 0.7 Charbonnier + 0.3 MSE loss. Epoch 15 is used for the strict matched sensitivity reported here.

3.3 Downstream classifier

The downstream probe is a ResNet-18 trained from scratch on 256 x 256 grayscale inputs, with its first convolution changed to one channel and its final layer changed to five classes. It uses class-balanced

weighted sampling, AdamW at 1e-3, cross-entropy loss, batch size 64, mixed precision, and 12 epochs. The selected classifier epoch maximizes validation Macro-F1; test boards are never used for selection. The stress evaluator adds deterministic Gaussian noise with sigma 25 on the 8-bit scale. Random, NumPy, PyTorch CPU, CUDA, sampler, data split, augmentation, and filename-hashed stress-noise seeds are controlled.

4 Evaluation and statistics

The corrected random stress replication uses seeds 42, 13, 37, 101, and 202 for denoiser training and downstream classifier training. We report mean, sample standard deviation, t-based 95% confidence intervals, paired seed effects, and exact sign-flip p-values. With n=5, these are effect-size estimates and not definitive confirmation. The earlier three-run table is retained only as an unverified pilot diagnostic because its launcher did not prove independent seed propagation.

The board-grouped protocol uses six locked folds. Each fold trains on four boards, selects the classifier checkpoint on a different validation board, and reports once on the held-out test board. Results are averaged with equal board weight. Because folds share training boards and all U-Net runs use seed 42, the six board effects are not six independent training-seed replicates. Their confidence intervals and sign-flip tests are descriptive.

5 Results

5.1 Corrected five-seed DnCNN replication

Table 2. Corrected seed-aware random stress replication (n=5).

Pipeline	Accuracy mean +/- SD	Accuracy 95% CI	Macro-F1 mean +/- SD	Macro-F1 95% CI
Raw	0.9328 +/- 0.0158	[0.9132, 0.9524]	0.8439 +/- 0.0164	[0.8234, 0.8643]
V4R light ECA	0.9254 +/- 0.0178	[0.9033, 0.9475]	0.8269 +/- 0.0227	[0.7987, 0.8551]
V5R aggressive ECA	0.9429 +/- 0.0144	[0.9250, 0.9608]	0.8589 +/- 0.0314	[0.8199, 0.8979]

V5R exceeded V4R on all five paired seeds. The V5R-minus-V4R estimate was +0.0175 Accuracy (95% CI [0.0060, 0.0291]) and +0.0319 Macro-F1 (95% CI [0.0085, 0.0554]); Cohen dz values were 1.88 and 1.69. The exact two-sided sign-flip p-value is nevertheless 0.0625 because five pairs give limited resolution. Relative to Raw, V5R changed Macro-F1 by only +0.0150 (95% CI [-0.0224, 0.0525]; 3/5 wins). Thus the corrected data reverse the old V4R-versus-V5R pilot ranking but do not establish a universal denoising advantage over Raw.

5.2 Locked transfer to unseen boards

Table 3. Equal-board six-fold transfer audit (seed 42).

Pipeline	Accuracy mean +/- SD	Macro-F1 mean +/- SD	Interpretation
Raw	0.8314 +/- 0.0615	0.4970 +/- 0.1032	Best mean Macro-F1
V4R light ECA	0.8294 +/- 0.0677	0.4395 +/- 0.1516	Heterogeneous transfer
V5R aggressive ECA	0.8321 +/- 0.1218	0.4681 +/- 0.1840	Near Raw accuracy; lower Macro-F1

The grouped audit estimates a different question from the random split: transfer to an unseen physical board. Raw has the strongest mean Macro-F1. V5R performs well on several wood boards but degrades on C02 and C03, and W02 remains a known generalization warning. Denoising therefore cannot be claimed to improve classification under every acquisition shift.

5.3 U-Net ECA x normalization ablation

Table 4. Strict epoch-15 U-Net comparison, equal-board means over six held-out boards.

Normalization	NoECA Acc.	ECA3 Acc.	NoECA Macro-F1	ECA3 Macro-F1	Paired Macro-F1 effect
GroupNorm	0.5940	0.7525	0.1833	0.3262	+0.1429 [0.0350, 0.2508]
BatchNorm	0.5348	0.6122	0.2650	0.2506	-0.0144 [-0.1088, 0.0799]

Under GroupNorm, ECA3 improves Macro-F1 on all six boards (exact sign-flip p=0.03125), with particularly large recoveries on C03 and W01. Under BatchNorm, the mean effect is -0.0144, ECA wins on only 2/6 boards, and p=0.75. The matched GN-minus-BN interaction estimate is +0.1573 Macro-F1 (95% CI [-0.0003, 0.3150]; 5/6 positive; p=0.0625). This suggests normalization-dependent ECA utility. It does not prove a population-level interaction because the denoisers and classifiers use one seed and the board folds are dependent.

5.4 Mechanistic audit

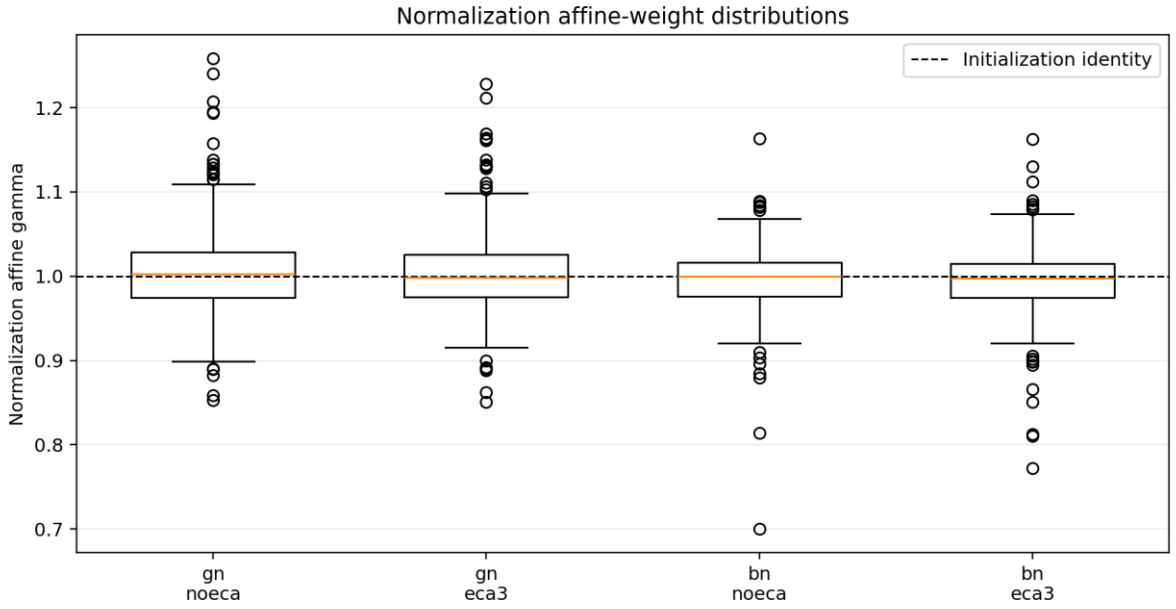


Figure 2. Affine gamma distributions in the four hash-locked U-Net checkpoints. The distributions remain centered near one and do not show a systematic inverse compensation pattern that would by itself absorb ECA.

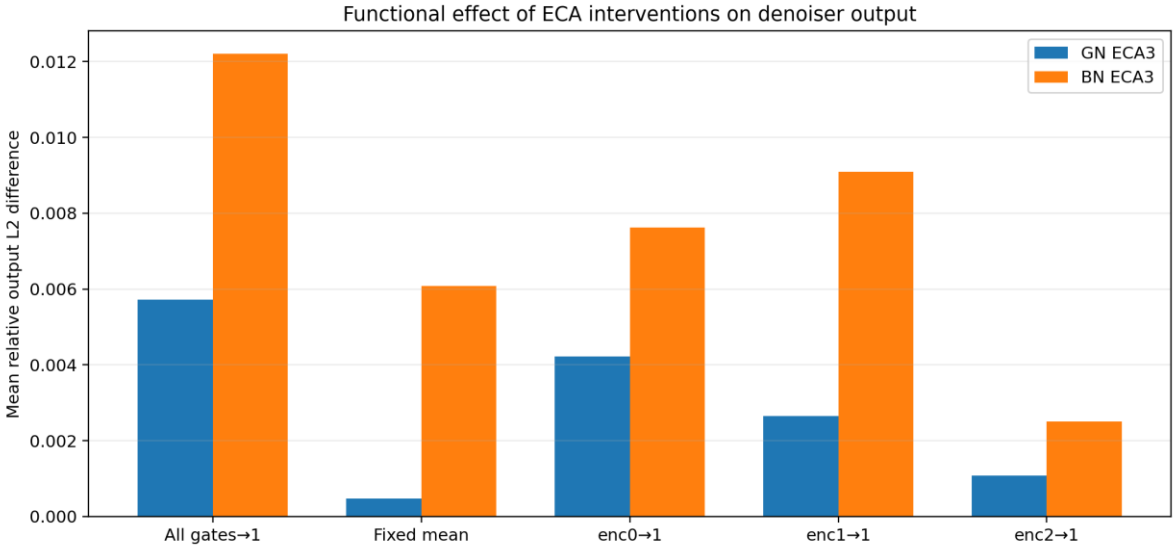


Figure 3. Mean relative denoiser-output change after inference-only ECA interventions on a disjoint audit set. BN-ECA is more dynamically active than GN-ECA; greater modulation is not equivalent to greater downstream utility.

Table 5. Disjoint 48-image audit after 48-image calibration; no training or fine-tuning.

Locked ECA model	Gates to 1: rel. L2	Fixed means: rel. L2	Fixed-means output corr.	Mechanistic reading
U-Net GN-ECA3	0.005720	0.000469	0.99865	Predominantly stable channel scaling
U-Net BN-ECA3	0.012200	0.006075	0.94340	More input-dependent modulation

The intervention audit rejects the simple mechanism that BatchNorm makes ECA invisible in the U-Net. Local ECA perturbations are attenuated along GroupNorm consumer paths but amplified along BatchNorm paths. The GN model is reproduced almost exactly by fixed calibration-mean channel multipliers, whereas the BN model retains greater input dependence. Normalization affine values remain centered close to one, and correlations between fixed gate means and downstream convolution channel norms are weak and inconsistent. Therefore the old DnCNN float-equivalence observation remains valid only for its specific pseudo-supervised BatchNorm residual configuration; it is not a universal theory of normalized attention.

5.5 Modern restoration negative control

An exploratory NAFNet-Tiny native-SCA control was evaluated on C01. Its output was nearly white (mean 0.9867, input-output correlation 0.0634, gradient retention 0.033) and downstream Accuracy/Macro-F1 was 0.0901/0.0778. This is not evidence that modern restoration backbones are intrinsically inferior. It is a negative control showing that the supplied averaged proxy can reward whitening that destroys ESPI class evidence. We did not expand this failed supervision contract to all folds.

6 Discussion

The revised evidence changes the story in three ways. First, aggressive V5R rather than light V4R is stronger in the corrected five-seed random stress protocol. Second, neither DnCNN variant consistently exceeds a noise-adapted Raw classifier on unseen boards. Third, ECA can strongly improve one U-Net normalization regime while providing no stable benefit in another. The appropriate contribution is therefore diagnostic, not architectural: ECA utility is conditional on supervision, normalization, topology, and the shift represented by the evaluation protocol.

Reconstruction and classification objectives are not interchangeable. Averaged targets suppress stochastic speckle but can also suppress weak fringe contrast. The NAFNet control and the near-white U-Net outputs show that optimizing the proxy can produce visually smooth, numerically plausible, but task-damaging mappings. Downstream validation must therefore accompany PSNR or loss when the scientific objective is classification.

The corrected experiment also illustrates a reproducibility lesson. A repeated-run table is not a seed study unless the seed reaches the split, model initialization, sampler, augmentation, stress noise, CUDA generator, and classifier training. We explicitly retire the earlier three-run result for inferential use. This corrected replication supersedes the previous three-run robustness summary for inferential purposes.

7 Limitations

- Five paired seeds improve the random stress estimate but remain underpowered for exact randomization tests; the reported intervals and effect sizes are suggestive.
- The U-Net 2 x 2 ablation uses one denoiser/classifier seed. Board folds share training boards and are not independent seed replicates.
- The grouped audit includes only six physical boards and strong class imbalance; per-class F1 remains heterogeneous, especially on C02, C03, and W02.
- Averaged targets are proxies rather than noise-free ground truth, and 243 cached training inputs are PseudoNoisy rather than real single shots.
- The mechanistic interventions are inference-only probes. Setting gates to one is not equivalent to retraining a NoECA model.
- The NAFNet-Tiny result is a single-board negative control and must not be reported as a general architecture ranking.

8 Conclusion

ECA is neither universally invisible nor universally useful in ESPI denoising. The corrected five-seed replication suggests that aggressive V5R is more robust than light V4R under the random stress protocol, while the six-board transfer audit shows no universal denoising advantage over Raw. In a residual U-Net, ECA improves GroupNorm Macro-F1 but not BatchNorm Macro-F1 at a matched epoch. Mechanistic interventions further show that BatchNorm does not simply erase ECA; it propagates more dynamic modulation that is not useful under the current supervision contract. The reliable conclusion is conditional: attention should be judged by its effective mapping, its interaction with normalization and supervision, and its downstream preservation of class-discriminative ESPI structure.

Artifact availability

Seed-aware orchestration, aggregation code, locked result tables, U-Net normalization analyses, and paper-ready revision inserts are available in the public repository: https://github.com/GeorgeSpy/ESPI-DnCNN-ECA. Raw acquisition data are not redistributed by this manuscript artifact.

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